

GRouNdGAN Top- k Causal-Graph Scaling Report

Auto-generated from `tf-k-experiment` run logs

September 22, 2026

1 Overview

This report analyses 15 completed runs in which the top- k GRN parameter was varied from 1 to 15. The number of imposed edges is used as the independent variable throughout the scaling analysis. The HVG count, TF count, and target-gene count were intended to remain fixed.

Observed dimensions. The logs show a small deviation from perfectly fixed dimensions: the $k = 1$ run has 2,997 genes and 194 TFs, while most later runs have 3,000 genes and 197 TFs. The observed values are reported rather than silently normalised.

2 GRN Characteristics

Table 1: GRN statistics by top- k setting. The independent variable is imposed-edge count.

Top- k	TFs	Targets	Genes	TF/Target (%)	Imposed Edges	Possible Edges	Density (%)
1	194	2803	2997	6.92	2,803	543,782	0.5
2	197	2803	3000	7.03	5,606	552,191	1.0
3	197	2803	3000	7.03	8,409	552,191	1.5
4	197	2803	3000	7.03	11,212	552,191	2.0
5	197	2803	3000	7.03	14,015	552,191	2.5
6	197	2803	3000	7.03	16,818	552,191	3.0
7	197	2803	3000	7.03	19,621	552,191	3.6
8	197	2803	3000	7.03	22,424	552,191	4.1
9	197	2803	3000	7.03	25,227	552,191	4.6
10	197	2803	3000	7.03	28,030	552,191	5.1
11	197	2803	3000	7.03	30,833	552,191	5.6
12	197	2803	3000	7.03	33,636	552,191	6.1
13	197	2803	3000	7.03	36,439	552,191	6.6
14	197	2803	3000	7.03	39,242	552,191	7.1
15	197	2803	3000	7.03	42,045	552,191	7.6

3 Training Step Times

Table 2: Average final-step times for the controller and causal generator.

Top- k	Imposed Edges	Controller (ms)	Generator (ms)
1	2,803	42	71
2	5,606	43	57
3	8,409	41	60
4	11,212	40	54
5	14,015	39	71
6	16,818	37	73
7	19,621	38	101
8	22,424	38	98
9	25,227	41	126
10	28,030	39	127
11	30,833	41	159
12	33,636	33	175
13	36,439	40	185
14	39,242	39	203
15	42,045	38	228

Table 3: Expected duration for 200,000 controller steps and 500,000 generator steps.

Top- k	Imposed Edges	Controller	Generator / Total
1	2,803	2.3 h	9.9 h / 12.2 h
2	5,606	2.4 h	7.9 h / 10.3 h
3	8,409	2.3 h	8.3 h / 10.6 h
4	11,212	2.2 h	7.5 h / 9.7 h
5	14,015	2.2 h	9.9 h / 12.0 h
6	16,818	2.1 h	10.1 h / 12.2 h
7	19,621	2.1 h	14.0 h / 16.1 h
8	22,424	2.1 h	13.6 h / 15.7 h
9	25,227	2.3 h	17.5 h / 19.8 h
10	28,030	2.2 h	17.6 h / 19.8 h
11	30,833	2.3 h	22.1 h / 24.4 h
12	33,636	1.8 h	24.3 h / 26.1 h
13	36,439	2.2 h	25.7 h / 27.9 h
14	39,242	2.2 h	28.2 h / 30.4 h
15	42,045	2.1 h	31.7 h / 33.8 h

4 Parameter Counts

Top- k	Edges	Generator	Critic	Labeler	Anti-Lab.	Gen. Opt.	Crit. Opt.	Labeler Opt.	Anti-Lab. Opt.
1	2,803	3.97M	3.73M	14.02M	14.02M	0.28M	11.18M	42.04M	42.04M
2	5,606	4.15M	3.73M	14.03M	14.03M	0.49M	11.19M	42.05M	42.05M
3	8,409	4.38M	3.73M	14.03M	14.03M	0.75M	11.19M	42.05M	42.05M
4	11,212	4.66M	3.73M	14.03M	14.03M	1.06M	11.19M	42.05M	42.05M
5	14,015	4.99M	3.73M	14.03M	14.03M	1.42M	11.19M	42.05M	42.05M
6	16,818	5.36M	3.73M	14.03M	14.03M	1.83M	11.19M	42.05M	42.05M
7	19,621	5.79M	3.73M	14.03M	14.03M	2.30M	11.19M	42.05M	42.05M
8	22,424	6.27M	3.73M	14.03M	14.03M	2.81M	11.19M	42.05M	42.05M
9	25,227	6.80M	3.73M	14.03M	14.03M	3.37M	11.19M	42.05M	42.05M
10	28,030	7.38M	3.73M	14.03M	14.03M	3.99M	11.19M	42.05M	42.05M
11	30,833	8.01M	3.73M	14.03M	14.03M	4.65M	11.19M	42.05M	42.05M
12	33,636	8.69M	3.73M	14.03M	14.03M	5.36M	11.19M	42.05M	42.05M
13	36,439	9.43M	3.73M	14.03M	14.03M	6.13M	11.19M	42.05M	42.05M
14	39,242	10.21M	3.73M	14.03M	14.03M	6.95M	11.19M	42.05M	42.05M
15	42,045	11.04M	3.73M	14.03M	14.03M	7.81M	11.19M	42.05M	42.05M

5 GPU Memory

Table 4: Peak GPU memory by training phase.

Top- k	Imposed Edges	Controller (MiB)	Generator (MiB)
1	2,803	819	1,919
2	5,606	795	2,113
3	8,409	795	2,091
4	11,212	1,309	2,897
5	14,015	1,143	3,327
6	16,818	795	3,699
7	19,621	1,143	4,865
8	22,424	1,309	5,971
9	25,227	1,309	7,005
10	28,030	795	7,643
11	30,833	1,309	9,425
12	33,636	1,309	10,799
13	36,439	749	12,289
14	39,242	1,143	13,703
15	42,045	749	15,075

6 Linear and Quadratic Fits

Let e denote the number of imposed edges. Ordinary least-squares fits over all completed runs give:

$$\widehat{M}_{\text{gen}}(e) = 0.347271 e - 932.5 \text{ MiB}, \quad R^2 = 0.9504. \quad (1)$$

$$\widehat{t}_{\text{gen}}(e) = 0.004435 e + 19.74 \text{ ms/step}, \quad R^2 = 0.9173. \quad (2)$$

$$\widehat{P}_{\text{gen},2}(e) = 0.003207968 e^2 + 36.156489 e + 3848734.0 \text{ parameters}, \quad R^2 = 1.0000. \quad (3)$$

$$\widehat{M}_{\text{gen},2}(e) = 0.000007201 e^2 + 0.024320 e + 1632.4 \text{ MiB}, \quad R^2 = 0.9977. \quad (4)$$

$$\widehat{t}_{\text{gen},2}(e) = 0.000000109 e^2 - 0.000437 e + 58.44 \text{ ms/step}, \quad R^2 = 0.9809. \quad (5)$$

The quadratic fits substantially improve in-sample fit quality, but their extra curvature should be validated on additional top- k values before extrapolation.

7 Graphs

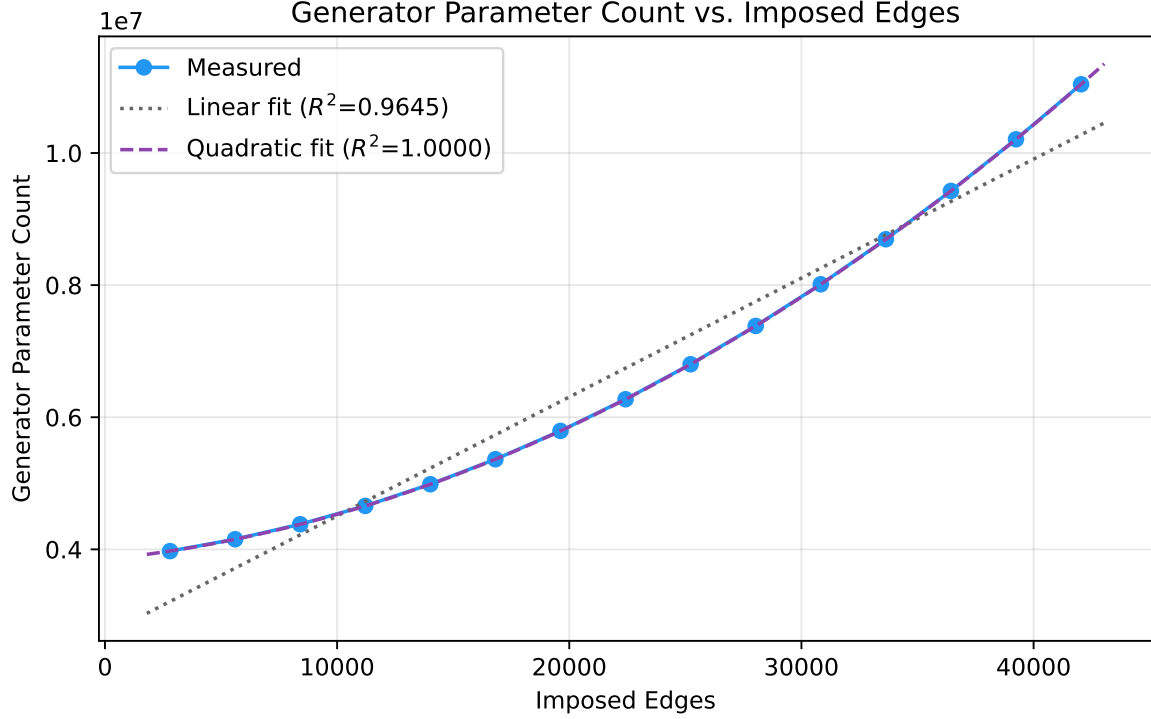


Figure 1: Generator parameter count versus imposed edges.

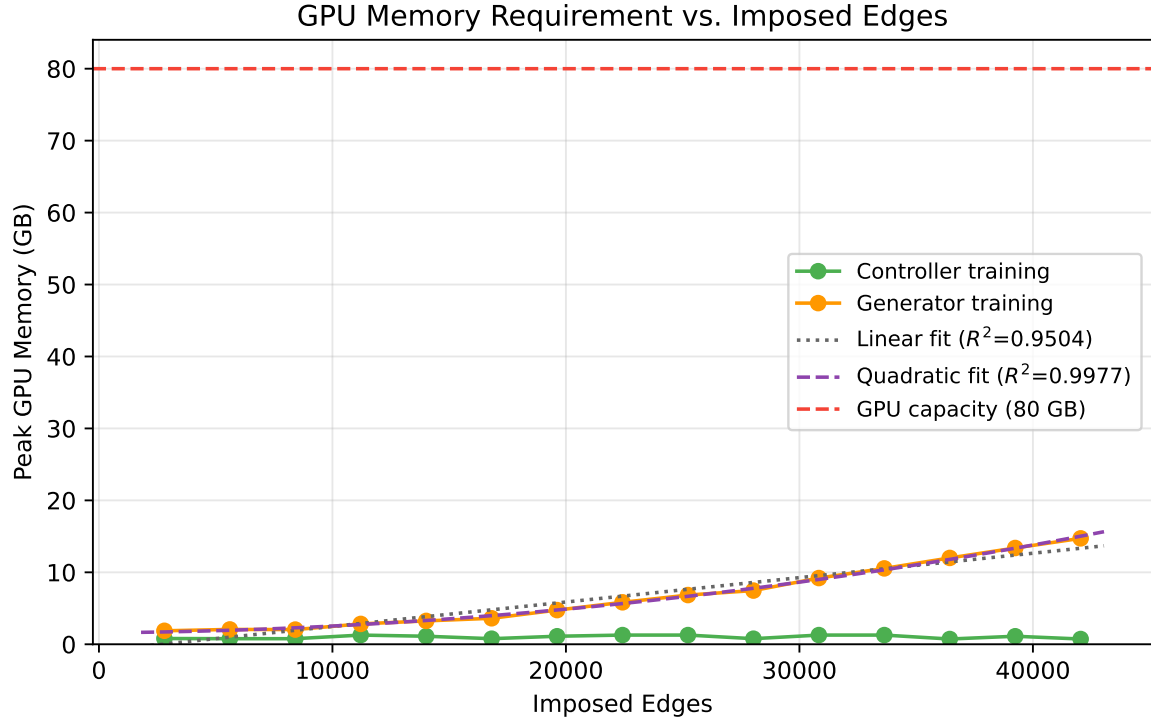


Figure 2: Peak GPU memory versus imposed edges; the dashed line is the 80 GB GPU capacity.

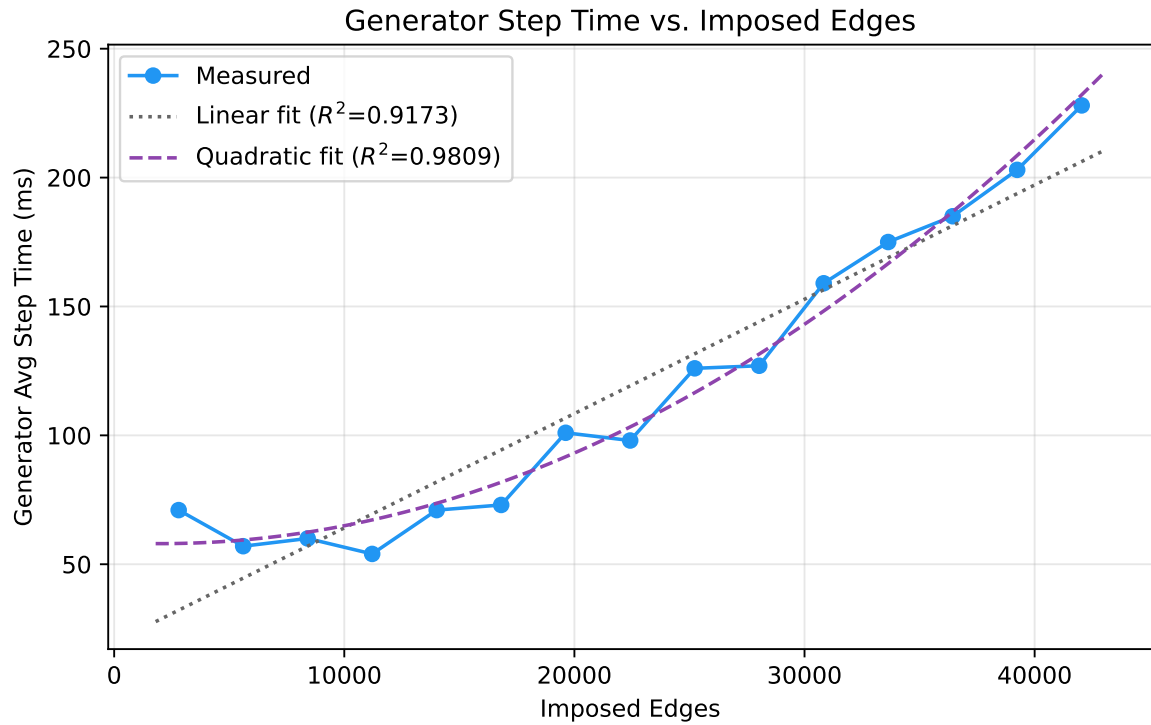


Figure 3: Causal-generator average step time versus imposed edges.

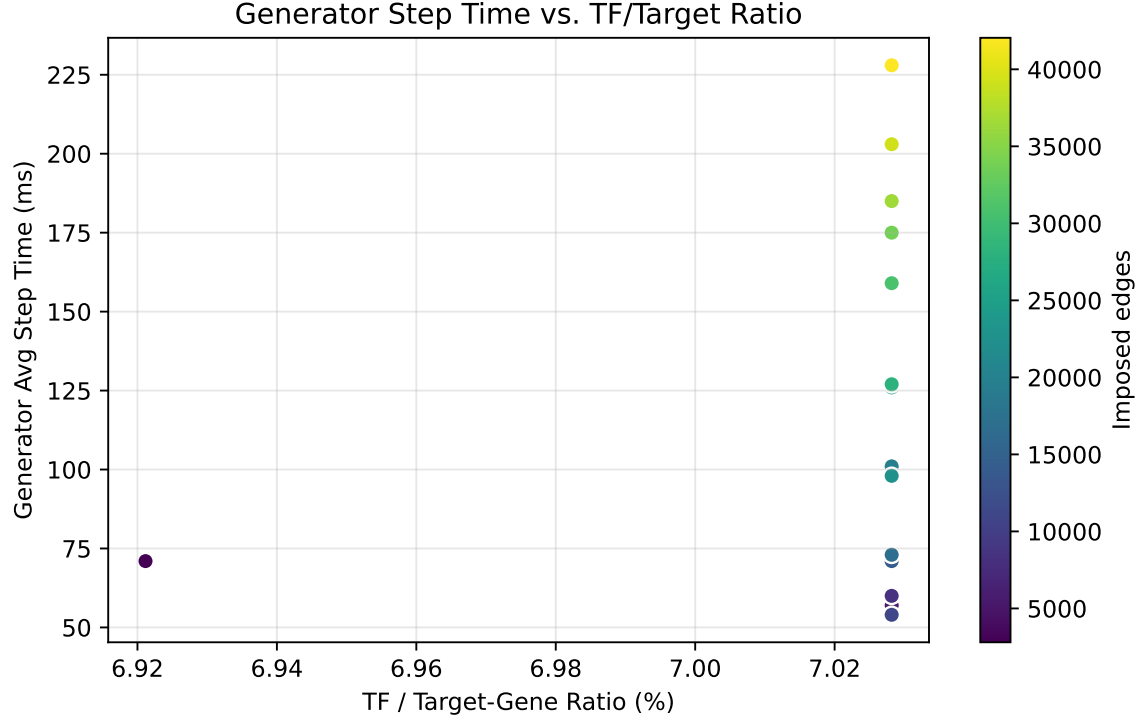


Figure 4: Causal-generator step time versus TF/target ratio; color indicates imposed-edge count.

8 Notes

- All included runs completed without CUDA out-of-memory errors.
- Generator parameter count scales approximately linearly with imposed edges (linear $R^2 = 0.9645$; quadratic $R^2 = 1.0000$).
- Quadratic fits improve the parameter-count R^2 from 0.9645 to 1.0000, the memory R^2 from 0.9504 to 0.9977, and the step-time R^2 from 0.9173 to 0.9809; this comparison is in-sample because all available top- k values were used for fitting.
- GPU memory is reported in MiB as measured by the monitoring CSV; the GPU capacity is 81,920 MiB.